

Course: Natural Language Processing

Title: Related Work Review and Literature Analysis



Submitted By: Junaid Ameer Khan

Roll NO: NUM-BSCS-2022-46

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Department of Computer Science
NAMAL University

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1. Introduction

In the contemporary landscape of health and wellness, personalized fitness guidance has become increasingly crucial. However, many existing solutions fall short in providing dynamic, context-aware, and comprehensive support. Users frequently perform exercises incorrectly, leading to suboptimal results and a heightened risk of injury. While computer vision (CV) offers a powerful means for real-time posture assessment, it often lacks the capacity to deliver nuanced, natural language explanations or tailored advice. Conversely, current chatbot systems, despite their proficiency in natural language interaction, tend to offer generic responses that fail to integrate user-specific workout data or real-time visual feedback. This project endeavors to bridge this significant gap by developing an innovative system that synergistically combines the precision of computer vision for accurate posture analysis with the intelligence of Retrieval-Augmented Generation (RAG) for personalized, context-aware, and actionable fitness guidance.

This literature review constitutes the second pivotal milestone of our semester project, building directly upon the problem definition established in Assignment 1. Its overarching objective is to meticulously survey and critically analyze the cutting-edge research pertinent to our proposed system. We aim to gain a profound understanding of how similar challenges have been addressed by the scientific community, to rigorously compare diverse methodologies, to evaluate various data strategies, and, crucially, to identify existing research gaps that our project seeks to fill. The review is systematically structured into three primary sections, reflecting the historical evolution and the eventual convergence of the relevant technological domains: (1) Traditional Machine Learning and Early Computer Vision Approaches, (2) Deep Learning and Advanced Computer Vision for Posture Correction, and (3) Transformer-Based and Retrieval-Augmented Generation (RAG) Methods. Each section will delve into specific research papers, dissecting their contributions, methodologies, datasets, evaluation metrics, key findings, and inherent limitations, thereby constructing a robust theoretical foundation for the subsequent implementation phase of our project.

2. Literature Review

2.1 Traditional Machine Learning and Early Computer Vision Approaches

The nascent stages of automated fitness monitoring were predominantly characterized by the application of traditional machine learning (ML) techniques, often augmented by rudimentary computer vision algorithms or sensor-based systems. These foundational approaches were instrumental in establishing the preliminary frameworks for understanding human movement and discerning exercise patterns. However, they frequently encountered limitations concerning accuracy, adaptability to diverse scenarios, and the richness and granularity of the feedback they could provide.

2.1.1 Early Computer Vision for Exercise Analysis

One of the seminal contributions in this domain was presented by Nagarkoti et al. in 2019 [1]. Their research focused on developing a real-time indoor workout analysis system designed to detect incorrect exercise forms. The methodology primarily involved the

application of classical computer vision algorithms, such as feature extraction (e.g., SIFT, HOG) and object detection, coupled with conventional machine learning classifiers (e.g., Support Vector Machines, Random Forests). The system processed video streams to identify key anatomical landmarks and analyze their trajectories, thereby determining the correctness of exercise execution. While the system demonstrated commendable accuracy in posture detection for a predefined and limited set of exercises, a significant drawback was the absence of semantic feedback. The system could effectively flag an error but was incapable of articulating *why* the error occurred or *how* to rectify it in natural language. This fundamental limitation underscores a critical gap that our proposed RAG-based system explicitly aims to address: the transformation of raw visual cues into intelligible, actionable linguistic advice that empowers users to understand and correct their form.

Complementing this, Khanal et al. (2022) [3] provided an exhaustive review of computer vision technologies as applied to physical exercise monitoring. Their comprehensive analysis meticulously traced the evolutionary trajectory from cumbersome sensor-based systems, which often suffered from issues of obtrusiveness, discomfort, and restricted mobility to the more flexible and non-invasive vision-based assessment methodologies. The authors highlighted that while vision-based methods offered substantial advantages in terms of user experience and data collection, the persistent challenge lay in the seamless integration of real-time corrective feedback in a user-friendly and pedagogically effective manner. The review unequivocally emphasized the imperative for systems that not only accurately detect errors but also provide intelligent, personalized, and contextually relevant guidance. This directly aligns with a core objective of our project: to move beyond mere error detection towards a holistic guidance system.

2.1.2 Traditional Machine Learning for Recommendation Systems

Beyond the realm of visual analysis, early personalized fitness systems frequently employed traditional machine learning paradigms for generating recommendations. Mahyari et al. (2022) [2] conducted research into recommendation systems that learned from expert behaviors within deep recommendation frameworks. Their approach ingeniously leveraged historical user data, including past workout logs and performance metrics, alongside expert-defined exercise sequences to generate highly personalized workout plans. By meticulously analyzing user profiles and their past performance, the system aimed to significantly enhance the relevance and efficacy of its recommendations. However, a primary constraint of this methodology was its inherent reliance on structured data logs and predefined rule sets, which rendered it less adaptable to dynamic, unstructured textual queries or spontaneous changes in user intent. The system struggled to effectively incorporate real-time, context-dependent information, a capability that is absolutely vital for delivering truly personalized and responsive guidance. This limitation starkly contrasts with our project's central focus on integrating real-time visual context with dynamic knowledge retrieval from a comprehensive knowledge base.

Toyonaga (2025) [4] further investigated the comparative effectiveness of content-based versus context-aware recommender systems in the promotion of physical activity. The study's findings indicated that systems capable of incorporating contextual information—such as the time of day, the user's current geographical location, or their recent activity

levels—demonstrated a statistically significant enhancement in user adherence and overall engagement. Nevertheless, this particular study predominantly focused on the analysis of structured activity logs and relied on a predetermined set of contextual features, rather than engaging in dynamic retrieval from a vast, unstructured knowledge base. The formidable challenge of seamlessly integrating disparate data sources (e.g., visual data, textual information, historical performance records) and subsequently providing coherent, personalized feedback remained a complex, multifaceted problem that traditional ML approaches could only partially address, thus leaving ample room for innovation.

2.1.3 Early Deep Learning Integration

With the nascent emergence of deep learning methodologies, a subset of studies began to cautiously integrate rudimentary neural network architectures into their systems. Sitapara et al. (2023) [5] introduced a Convolutional Neural Network (CNN)-based approach specifically designed for real-time exercise posture correction. Their pioneering work unequivocally demonstrated that even relatively lightweight CNN models, when meticulously trained on specialized gym datasets, could substantially outperform conventional machine learning methods in highly specific posture recognition tasks. The system exhibited the capability to accurately identify common errors in fundamental exercises such as squats and push-ups. However, the authors candidly acknowledged significant scalability issues, particularly when attempting to deploy such computationally intensive models on resource-constrained edge devices or when striving to generalize their applicability across an expansive spectrum of exercises and diverse body types. Furthermore, the feedback provided by their system was often confined to simplistic classifications (e.g., “correct” or “incorrect”) without offering detailed, explanatory natural language insights, thereby highlighting the pressing need for more sophisticated and deeply integrated Natural Language Processing (NLP) capabilities.

2.2 Deep Learning and Advanced Computer Vision for Posture Correction

The profound advancements in deep learning, particularly within the domain of computer vision, precipitated a revolutionary transformation in human pose estimation and action recognition. This section meticulously examines the sophisticated CV techniques that have enabled the development of more accurate, robust, and real-time exercise posture correction systems, which collectively form a critical and indispensable component of our proposed fitness guidance system.

2.2.1 Real-Time Pose Estimation and Error Classification

Building upon the foundational efforts discussed previously, Kotte et al. (2023) [6] proposed a groundbreaking strategy that harnessed advanced computer vision methods to provide real-time feedback on posture during gym exercises. Their system employed highly sophisticated pose estimation models, such as Open Pose or Alpha Pose, to precisely track key anatomical points (e.g., joints, limbs) and meticulously analyze their dynamic movements. By rigorously comparing these observed movements against a meticulously curated database of expert-defined correct forms, the system was capable of classifying specific errors with a high degree of accuracy (e.g., identifying knee valgus during squats,

detecting insufficient depth in push-ups, or recognizing improper back alignment in deadlifts). While this represented a substantial improvement over earlier systems, the authors noted that the system still faced considerable challenges in complex multi-user environments and often necessitated meticulous calibration for optimal performance. Crucially, the feedback, although more specific and granular, remained primarily visual (e.g., graphical overlays on video feeds) or numerical (e.g., angle measurements), still lacking the rich, explanatory power and nuanced understanding inherent in natural language. This further reinforced the need for robust NLP integration.

Further specializing in the demanding field of powerlifting, Ko et al. (2024) [7] introduced an innovative real-time AI posture correction service specifically tailored for three major powerlifting exercises: the bench press, the squat, and the deadlift. Their meticulously designed methodology involved the deployment of advanced computer vision models, often based on deep neural networks, to accurately track intricate joint angles and precise body positions throughout these complex, high-intensity movements. By continuously comparing these real-time biomechanical metrics against a rigorously defined set of biomechanically correct forms, the system delivered immediate and actionable corrective feedback to users. The authors reported impressive F1-scores on specialized powerlifting datasets, unequivocally demonstrating the efficacy and precision of their approach for these specific, high-risk exercises. However, the inherent specialization of the system implied that it might not generalize effectively to a broader spectrum of fitness activities without extensive and costly retraining. Furthermore, the nature of the feedback provided was still largely prescriptive (e.g., “lower your hips more,” “maintain a neutral spine,” “keep your elbows tucked in”) rather than conversational or contextually rich, thereby limiting its pedagogical utility and user engagement.

In a similar vein, Rao et al. (2025) [8] concentrated their efforts on refining real-time posture correction, specifically for squat exercises. Leveraging cutting-edge deep learning techniques, their system was designed to assist beginners in assiduously improving their form by identifying even subtle deviations from the ideal posture. The system ingeniously utilized a hybrid architecture combining convolutional neural networks (CNNs) for spatial feature extraction and recurrent neural networks (RNNs) for analyzing temporal sequences of movements, thereby enabling it to provide more nuanced and dynamic feedback than static pose estimation alone. While demonstrably effective for a single exercise, the broader applicability to a comprehensive fitness regimen and the seamless integration with an expansive, comprehensive knowledge base remained significant open challenges. The feedback mechanism was predominantly visual overlays or concise textual cues, still lacking the depth and interactive nature of a natural language dialogue system.

2.2.2 Personalized Recommendations with Deep Learning

Beyond the immediate realm of real-time posture correction, deep learning also played a pivotal role in the development of more sophisticated personalized recommendation systems. Chen et al. (2025) [9] presented a robust machine learning framework meticulously designed to generate personalized fitness recommendations that were meticulously aligned with national health goals. Their holistic approach integrated a diverse array of data points, including user biometrics (e.g., age, weight, height), historical

workout data, and explicitly stated health objectives, to construct highly tailored exercise plans. While the system proved successful in generating personalized plans, its operation was primarily confined to a planning level, conspicuously lacking a real-time, interactive conversational interface for guiding users *during* their actual workouts. The recommendations generated were essentially static outputs rather than dynamic, context-aware advice, thereby highlighting the critical necessity for robust NLP integration to render the system truly interactive, responsive, and engaging for the end-user.

2.2.3 Advanced Pose Estimation and Performance Analysis

Researchers at ACM (2023) [10] made significant strides by demonstrating the formidable capability of human pose estimation to accurately determine the correctness of bicep curl technique in real-time. Their system ingeniously utilized readily available, off-the-shelf pose estimation libraries (e.g., MediaPipe, OpenPose) and subsequently applied custom-developed logic to meticulously evaluate parameters such as the range of motion, the velocity of execution, and the overall form of bicep curls. A crucial and insightful finding from their study was the profound impact of environmental factors, particularly varying lighting conditions, on the accuracy and overall reliability of the pose estimation. This observation starkly underscores the practical challenges inherent in deploying CV-based systems in diverse and uncontrolled user environments. The feedback provided was immediate but predominantly binary (e.g., “correct” or “incorrect”) or quantitative (e.g., “elbows too far out,” “insufficient wrist rotation”), conspicuously lacking deeper linguistic explanations that could facilitate genuine user understanding and learning.

Further propelling the field forward, Nature (2025) [11] reported on an significantly improved Convolutional Neural Network (CNN) framework specifically engineered for precise exercise posture recognition. This advanced framework not only excelled at recognizing a wide array of postures but also seamlessly integrated health indicator prediction, thereby offering a more holistic and comprehensive assessment of exercise performance. The enhanced CNN architecture was meticulously designed to surmount some of the inherent limitations of earlier models, providing more robust and accurate recognition capabilities across a diverse range of users and various exercise variations. However, the substantial computational cost associated with such sophisticated models remained a formidable challenge, particularly for their seamless deployment on resource-constrained edge devices or in real-time mobile applications. The paper implicitly advocated for the development of more efficient model architectures and potentially distributed computing paradigms to render such powerful systems widely accessible and practically viable.

2.3 Transformer-Based and Retrieval-Augmented Generation (RAG) Methods

The advent of Transformer architectures and the subsequent development of Retrieval-Augmented Generation (RAG) have heralded a new epoch for natural language processing. These innovations have enabled the creation of more sophisticated, context-aware, and knowledge-grounded conversational AI systems. This section meticulously explores how these advanced NLP techniques are being synergistically integrated with computer vision to forge intelligent and highly effective fitness guidance systems.

2.3.1 LLMs for Exercise Retrieval and Guidance

Large Language Models (LLMs) have unequivocally demonstrated extraordinary capabilities in comprehending and generating human-like text with remarkable fluency and coherence. Xu et al. (2024) [12] meticulously explored the application of LLMs for augmented exercise retrieval, particularly within the pedagogical context of online language learning. While their primary research domain was language education, the underlying methodology possesses profound relevance to the fitness domain. They conclusively demonstrated that semantic search, powerfully driven by LLM embeddings, could effectively and accurately match ambiguous user queries to highly relevant educational content, even in a challenging zero-shot setting. This capability is absolutely pivotal for our project, where users might pose open-ended, nuanced questions about exercises, necessitating the system to retrieve precise and contextually appropriate information from an expansive fitness knowledge base. A notable limitation identified in their study was the potential for high latency in real-time interactive scenarios, especially when dealing with very large models, thereby suggesting the critical need for highly optimized retrieval mechanisms and efficient model serving strategies.

Recognizing the immense potential of LLMs for personalized health interventions, IEEE (2025) [13] reported on the development of a dedicated RAG agent specifically designed for personalized fitness and dietary guidance. This sophisticated system was engineered to process natural language voice input, accurately interpret complex user intent, retrieve pertinent information from a meticulously curated knowledge base, and subsequently generate highly coherent and synthesized speech responses. The seamless integration of voice input and synthesized speech significantly augmented the user experience, rendering the interaction more natural, intuitive, and hands-free. The RAG architecture played an indispensable role in grounding the LLM's responses in verifiable factual information, thereby substantially mitigating the inherent risk of hallucinations commonly observed in standalone generative models. However, the authors highlighted the considerable resource intensiveness associated with maintaining and querying a large, dynamic knowledge base, coupled with the substantial computational demands of LLMs, which collectively posed significant challenges for achieving widespread scalability and practical deployment.

Further exemplifying the remarkable versatility of RAG, IEEE (2024) [14] investigated its application in the development of educational chatbots tailored for sports. Their system adeptly utilized RAG to extract highly specific information from specialized sports repositories, thereby furnishing accurate and contextually relevant answers to a diverse range of user queries. This innovative approach proved highly effective in mitigating the pervasive problem of LLM hallucinations by ensuring that all generated responses were rigorously backed by verifiable retrieved facts. The study emphatically underscored the paramount importance of a meticulously curated and continuously updated knowledge base for optimizing the performance of any RAG system. While demonstrably successful in factual question-answering, the system primarily focused on textual interactions and did not explicitly integrate real-time visual feedback, which remains a cornerstone of our proposed comprehensive fitness guidance system.

2.3.2 Knowledge-Grounded RAG for Personalized Advice

To further augment the reliability, accuracy, and depth of personalized recommendations, IEEE (2025) [15] introduced a pioneering Graph-Based RAG approach specifically for dietary guidance. This innovative methodology ingeniously combined the formidable power of LLMs with the structured precision of knowledge graphs (KGs). By meticulously representing dietary information, nutritional facts, and user preferences as interconnected nodes and edges within a graph structure, the system was capable of executing more sophisticated retrieval operations and complex reasoning processes. The authors conclusively demonstrated that the synergistic integration of KGs with RAG yielded significantly more reliable and coherent recommendations when compared to systems relying exclusively on vector-based retrieval mechanisms. While the initial setup and ongoing maintenance of such a graph-based system can be inherently complex, this approach offers a highly promising direction for ensuring the factual accuracy, logical consistency, and overall trustworthiness of personalized fitness and dietary advice. The formidable challenge now lies in extending this sophisticated framework to seamlessly incorporate real-time visual data and dynamic user interactions, thereby creating a truly multimodal and adaptive guidance system.

Lai et al. (2025) [16] conducted a rigorous systematic review of the diverse applications of Large Language Models in the critical domain of exercise rehabilitation. Their comprehensive review acknowledged that while LLMs exhibit exceptional prowess in generating varied and coherent text, their direct application in sensitive fields such as medical or rehabilitation advice necessitates scrupulous grounding and validation. They unequivocally concluded that RAG is an indispensable component for ensuring the medical accuracy, safety, and ethical integrity of LLM-generated recommendations in exercise rehabilitation. The review astutely identified existing research gaps, particularly emphasizing the urgent need for LLMs to integrate real-time physiological data and highly personalized patient histories to provide truly effective and bespoke guidance. This crucial insight strongly reinforces our project's central hypothesis: that the synergistic combination of Computer Vision (CV) and RAG is absolutely essential for delivering comprehensive, safe, and effective fitness guidance.

2.3.3 Multimodal RAG and Context-Aware Systems

Pushing the frontiers of RAG capabilities, ArXiv (2025) [17] introduced RAG-HAR (Retrieval Augmented Generation-based Human Activity Recognition). This innovative framework was specifically designed to provide personalized coaching by intelligently retrieving similar activity patterns from an extensive knowledge base, even in a challenging training-free setting. By effectively leveraging the RAG paradigm, the system could accurately interpret complex human activities and subsequently offer highly context-aware feedback. While its primary focus was on robust activity recognition, its integrated RAG component is profoundly relevant for generating personalized guidance based on observed actions and historical data. The concept of a "training-free" RAG system is particularly attractive due to its inherent adaptability across a diverse range of exercises and varying user needs, offering a flexible and scalable solution.

In the specialized realm of sports coaching, IEEE (2025) [18] meticulously explored FC-RAG (Football Coaching with Multimodal Retrieval-Augmented Generation). This groundbreaking work emphatically underscored the critical necessity for multimodal retrieval, seamlessly integrating both video analysis (for visual cues) and textual data (for strategic insights), to provide holistic and comprehensive guidance to elite athletes. For instance, the system possessed the capability to meticulously analyze a player’s movement patterns extracted from video footage and then intelligently combine this visual information with textual coaching strategies to formulate and deliver comprehensive feedback. This sophisticated multimodal approach is directly analogous to our project’s ambitious goal of combining high-precision computer vision (for real-time posture assessment) with advanced natural language processing (for intelligent, explanatory guidance). The inherent complexity of integrating, synchronizing, and coherently interpreting diverse multimodal data streams remains a significant and ongoing research challenge.

Hindi et al.(2025) [19] dedicated their research to enhancing the precision and interpretability of RAG systems. This particular focus is of paramount importance for fitness applications, where inaccurate, ambiguous, or poorly explained advice can lead to severe injury, user frustration, or demotivation. Their research aimed to develop novel RAG architectures that not only efficiently retrieve highly relevant information but also present it in an exceptionally clear, understandable, and verifiable manner. Improved interpretability is crucial for building user trust and fostering a deeper understanding of the generated advice, thereby promoting better adherence and learning. The invaluable findings from this study are vital for meticulously designing a RAG component that provides transparent, actionable, and trustworthy fitness guidance, a key differentiator for our system.

Finally, ResearchSquare (2025) [20] presented the design and rigorous evaluation of a context-aware multimodal recommendation and Question-Answering (QA) system that explicitly and synergistically integrates computer vision and RAG. This pioneering system directly addresses the core, multifaceted challenge of our project: the seamless combination of visual posture data with extensive textual expert knowledge to deliver intelligent, personalized, and real-time feedback. Their seminal work conclusively demonstrated the practical feasibility and immense potential of such an integrated system, unequivocally showcasing its capacity to offer real-time, context-aware automation guidance. While still in its nascent stages of development, this research provides a robust proof-of-concept and a compelling vision for the viability and profound impact of our proposed “Context-Aware Personalized Fitness Guidance System,” validating our architectural choices and research direction.

3. Comparative Literature Table

Sr. No	Ref	Year	Methodology	Dataset/Domain	Key Findings/Contributions	Limitations/Gaps
1	[1]	2019	Traditional CV + ML Classifiers	Indoor Workout	High accuracy in posture detection	Lacked semantic,

				Videos	for predefined exercises.	natural language feedback for error correction.
2	[2]	2022	Deep Recommendation Systems	User Profiles, Expert-defined Exercise Data	Improved recommendation relevance by learning from expert behaviors.	Relied on structured data; limited ability to process unstructured textual queries.
3	[3]	2022	CV Technology Review	Various (Sensor-based to Vision-based systems)	Highlighted the transition to more flexible vision-based systems for monitoring.	Integration of real-time NLP feedback for personalized guidance remained a significant gap.
4	[4]	2025	Context-Aware Recommender Systems	Activity Logs, Contextual Features	Enhanced user adherence and engagement by incorporating contextual information.	Primarily used structured logs; did not involve dynamic knowledge retrieval from unstructured sources.
5	[5]	2023	CNN-based Posture Correction	Specialized Gym Datasets	Outperformed traditional ML for specific posture recognition tasks.	Scalability issues for diverse exercises; feedback limited to simple classifications.
6	[6]	2023	Advanced CV (Pose Estimation)	Gym Exercise Videos	Real-time error classification using sophisticated pose estimation models.	Challenges in multi-user environments; feedback was visual/numerical, not natural language.

7	[7]	2024	AI Posture Correction (CV)	Powerlifting Datasets	High F1-scores for specific powerlifting exercises with immediate feedback.	Specialization limited generalizability; feedback was prescriptive, not conversational.
8	[8]	2025	Deep Learning for Squat Correction	Squat Exercise Videos	Provided nuanced feedback for a single exercise using CNNs and RNNs.	Limited to a single exercise; integration with comprehensive knowledge base was an open challenge.
9	[9]	2025	ML Framework for Recommendations	User Biometrics, Workout History, Health Goals	Generated personalized exercise plans aligned with health objectives.	Lacked a real-time, interactive conversational interface for in-workout guidance.
10	[10]	2023	Human Pose Estimation	Bicep Curl Videos	Determined correctness of bicep curl technique in real-time.	Accuracy was sensitive to environmental factors, especially lighting conditions.
11	[11]	2025	Improved CNN for Posture Recognition	Health Indicator Datasets	Precise posture recognition integrated with health indicator prediction.	High computational cost for real-time mobile deployment.
12	[12]	2024	LLM Augmented Exercise Retrieval	Educational Content	Effective semantic search for ambiguous user queries in a zero-shot setting.	Potential for high latency in real-time interactions with very large models.

13	[13]	2025	RAG Agent (Voice Input/Output)	Fitness/Diet Knowledge Base	Voice-integrated, grounded responses for personalized fitness/diet guidance.	Resource intensiveness for knowledge base maintenance and LLM computation.
14	[14]	2024	RAG Chatbot for Sports	Specialized Sports Repositories	Reduced LLM hallucinations by grounding responses in retrieved facts.	Primarily focused on factual question-answering; lacked real-time visual integration.
15	[15]	2025	Graph-Based RAG + LLMs	Dietary Knowledge Graphs	Provided more reliable and coherent recommendations using structured KGs.	Complex setup and maintenance; challenge in integrating dynamic visual data.
16	[16]	2025	LLM Review (Exercise Rehabilitation)	Medical/Rehab Literature	Concluded RAG is indispensable for medical accuracy in LLM-generated advice.	Identified need for real-time physiological data and personalized patient histories.
17	[17]	2025	RAG-HAR (Activity Recognition)	Human Activity Datasets	Personalized coaching by retrieving similar activity patterns in a training-free setting.	Primarily focused on activity recognition; not a full interactive guidance system.
18	[18]	2025	Multimodal RAG (Video + Text)	Football Coaching Data	Holistic guidance for athletes by integrating video analysis and textual strategies.	Complexity of integrating, synchronizing, and interpreting diverse

						multimodal data streams.
19	[19]	2025	Precision and Interpretability of RAG	General LLM Applications	Enhanced precision and interpretability of RAG systems for better trust and understanding.	General focus; specific application to fitness guidance requires further adaptation.
20	[20]	2025	Multimodal QA (CV + RAG)	Integrated CV/RAG Datasets	Demonstrated feasibility of integrated CV+RAG for context-aware automation guidance.	Still in early stages; requires further development for robust, real-world deployment.

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5. Plagiarism and AI Usage Reports

5.1 Plagiarism Report Statement

The content presented in this assignment is the product of original synthesis, critical analysis, and intellectual effort applied to the selected research papers. All external ideas, methodologies, data, and direct quotations have been meticulously cited and attributed using the IEEE referencing style, ensuring full academic integrity. A formal plagiarism check has been diligently conducted, and the resulting similarity index is confirmed to be well within the permissible limits stipulated by Namal University's academic policies. This report affirms that the work is free from unauthorized appropriation of others' work and adheres to the highest standards of scholarly conduct.

5.2 AI Usage Report Statement

Artificial Intelligence tools were judiciously employed in the preparation of this assignment for the following specific and transparent purposes:

- **Literature Search and Discovery:** AI-powered search engines and academic databases were utilized to efficiently identify and filter high-quality, peer-reviewed research papers from reputable sources such as IEEE, ACM, Springer, and Nature, directly relevant to the project's scope.
- **Content Structuring and Drafting Assistance:** AI models provided assistance in structuring the comprehensive literature review, organizing the technical paragraphs, and formatting the comparative table. This aid focused on enhancing clarity, logical flow, and adherence to professional academic standards.
- **Grammar, Style, and Cohesion Enhancement:** AI tools were employed to refine the prose, correct grammatical errors, improve sentence structure, and ensure that the writing consistently maintained a professional, academic tone. This included checking for stylistic consistency and overall textual cohesion.

It is imperative to state that all AI-generated suggestions, drafts, and content were subjected to rigorous critical review, thorough editing, and meticulous verification by the human authors. The ultimate responsibility for the accuracy, technical correctness, and academic integrity of the final submission rests solely with the authors. AI was used as a productivity enhancement tool, not as a substitute for original thought or scholarly diligence.